

Homework 3

Advanced Models and Methods for Water Security

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1 Steady-State Simulation in a Heterogeneous Confined Aquifer

The objective of this study was to perform a steady-state simulation of flow in a heterogeneous confined aquifer influenced by multiple extraction wells. The study analyzes the hydraulic head distribution and evaluates the discrepancies between numerical outcomes (MODFLOW) and analytical solutions (Thiem equations).

1.1 Model Configuration and Methodology

The simulation was developed using Flopy, utilizing the code structure from Tutorial 2 as a baseline. The original homogeneous model was modified to introduce heterogeneity and accommodate a three-well system.

Well Configuration:

Working in section 5 of the code, a third well (Well 3) was introduced to the domain, and the positions and pumping rates of the two pre-existing wells were adjusted. All wells were assigned negative pumping rates to simulate extraction. The specific configuration for the three wells is detailed in Table .

Well ID	Row	Column	Pumping Rate (m^3/d)	Type
Well 1	20	17	-75	Extraction
Well 2	9	19	-40	Extraction
Well 3	15	3	-60	Extraction

Table 1: Well configuration

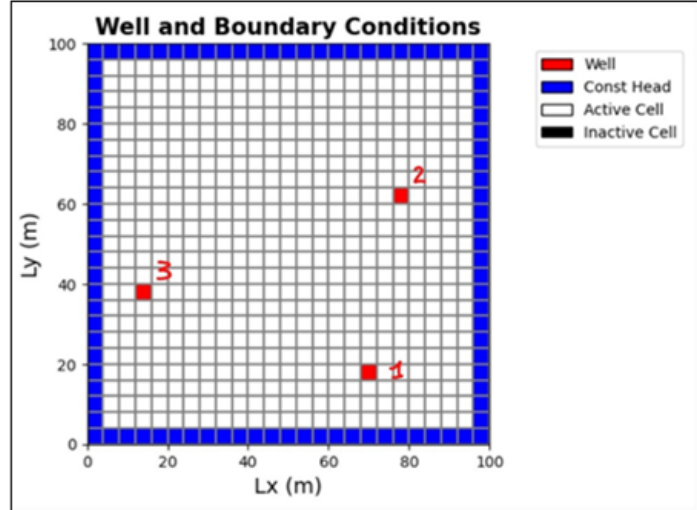


Figure 1: Plot of the grid showing the position of the 3 wells.

Heterogeneity Setup:

To simulate a heterogeneous matrix, the hydraulic conductivity (hk) was defined as a spatially variable matrix matching the grid dimensions ($nrow \times ncol$). The domain was divided into two distinct zones to represent different materials:

- **Left Zone (Columns 0–25):** Modeled as *Sand* with high hydraulic conductivity ($hks = 20.0 \text{ m/d}$), representing a zone where water flows easily.
- **Right Zone (Columns 25–50):** Modeled as *Clay* with low hydraulic conductivity ($hk = 1.0 \text{ m/d}$), representing a zone of high resistance to flow.

Vertical conductivity (vk) was maintained at default software values as it does not significantly impact this 2D horizontal flow analysis.

1.2 Results and Observations

The steady-state simulation yielded distinct hydraulic head distributions driven by the contrasting conductivity values and boundary conditions.

1.2.1 Observation 1: Boundary Effects on Well 3

Well 3 is located in the high-conductivity sand zone (Column 3). Theoretically, high conductivity suggests a wide, shallow cone of depression. However, the results show a steep hydraulic gradient between the well and the left boundary.

This phenomenon is caused by the proximity of Well 3 to the Constant Head Boundary (set at $h = 10 \text{ m}$ at Column 0).

- **Mechanism:** The well lowers the head to approximately 8m while the boundary remains fixed at 10m only 3 cells away.
- **Visual Interpretation:** The flow lines (head contours) are packed tightly between the well and the edge.

In hydraulics, tightly packed contours indicate a steep gradient, signifying that water is flowing rapidly from the boundary to recharge the well. The boundary effectively "feeds" the well, preventing a deep drawdown despite the pumping.

1.2.2 Observation 2: Drawdown in Sand vs. Clay

A clear contrast is observed between the drawdown cones in the two zones (Figure ??):

- **In the Clay (Right):** A deep, steep cone of depression is visible (represented by a "dark hole" or very low head values in the plot).
- **In the Sand (Left):** The drawdown is shallow and subtle.

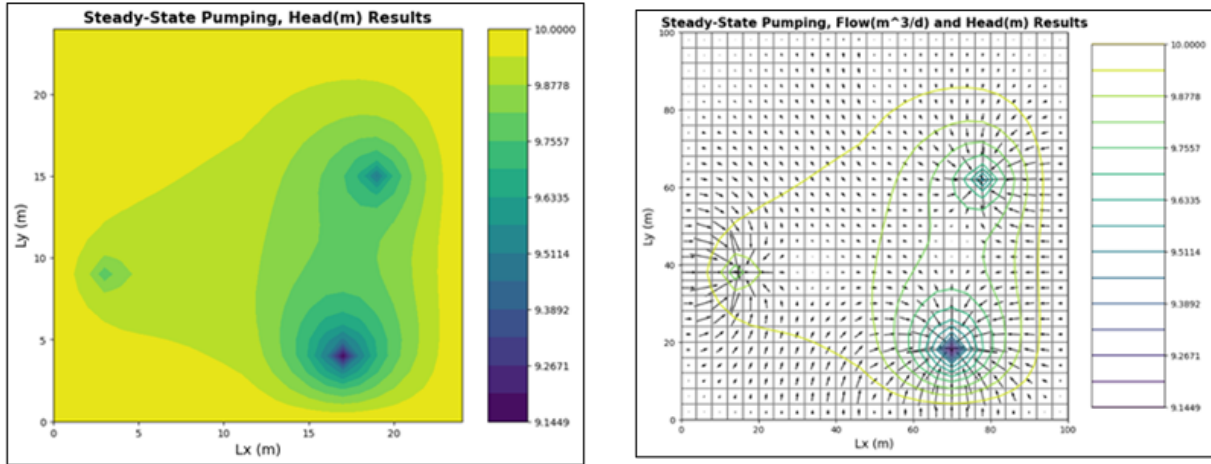


Figure 2: Steady-state hydraulic head distribution. Note the deep cone of depression in the low-conductivity clay zone (right) compared to the shallower drawdown in the high-conductivity sand zone (left).

Physical Explanation:

This disparity is explained by Darcy's Law:

$$Q = K \cdot i \cdot A \quad (1)$$

Where Q is the discharge (pumping rate), K is hydraulic conductivity, and i is the hydraulic gradient. Since both zones are subject to extraction (negative Q), the aquifer must generate a hydraulic gradient to move water toward the well.

- **High Efficiency (Sand):** With high K , the aquifer offers low resistance. Only a small gradient (i), and therefore minimal drawdown, is required to sustain the flow Q .
- **Low Efficiency (Clay):** With low K , the resistance to flow is high. To maintain the same flux Q , the system must generate a massive pressure drop (high gradient i), resulting in a deep cone of depression.

1.3 Comparison: Numerical vs. Analytical Solutions

The simulation results were compared against an analytical solution derived using the Thiem equation.

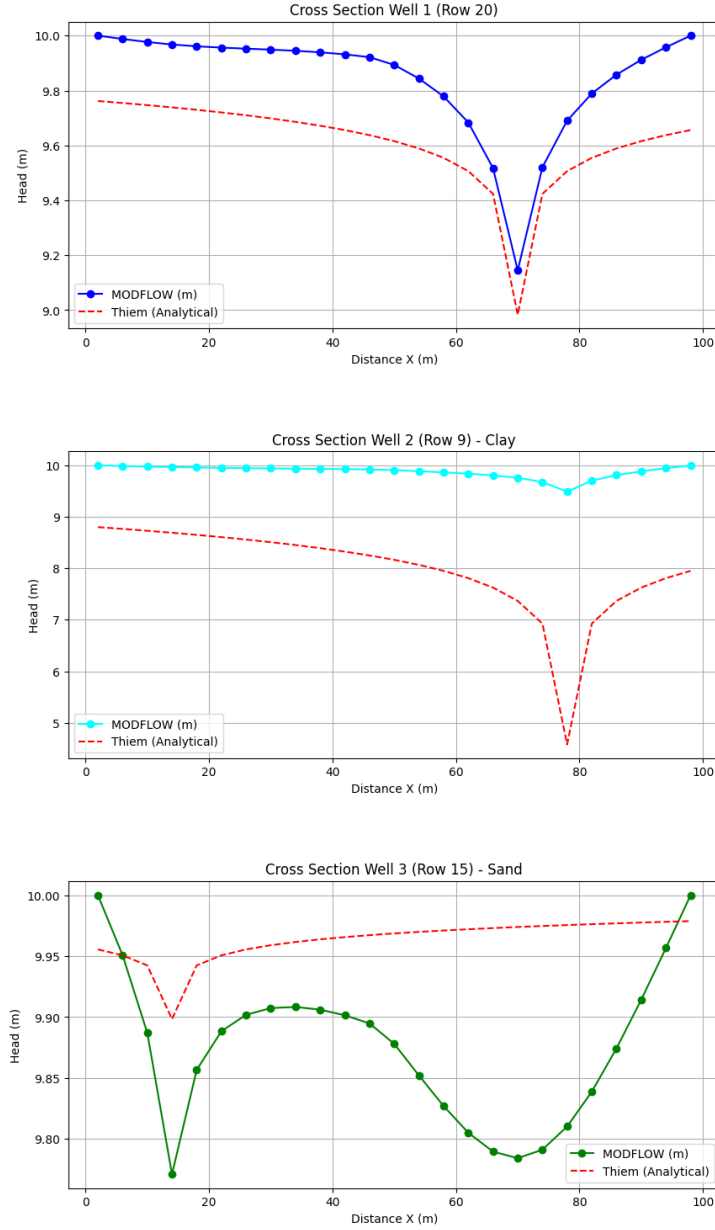


Figure 3: Comparison of drawdown profiles between the MODFLOW numerical solution (Blue, light blue and green) and the analytical Thiem solution (Red).

1.3.1 Analysis of Discrepancies

As seen in Figure 3, there are significant discrepancies between the two methods:

- **The Analytical Solution (Red):** Represents an idealization. It assumes an infinite, homogeneous, radial system. It predicts a deeper drawdown at the well because it is not limited by grid cell resolution, and it drifts at the edges because it does not account for the constant head boundaries.
- **The Numerical Solution (Blue, light blue and green):** Provides a more accurate representation of the specific domain geometry. It accounts for the linear boundaries (at $X=0$ and $X=100$) and the spatial heterogeneity.

While the analytical formula provides a theoretical baseline, it cannot predict the asymmetrical drawdown cones and boundary-controlled heads observed in the numerical model. The modflow model correctly captures the "flattening" of the cone near the recharge boundary and the steepening of the cone in the low-conductivity zone, highlighting the necessity of numerical modeling for complex, heterogeneous systems.

2 Unsteady-state simulation of flow with different flow rates over various time periods

The goal of this section is to perform a transient (unsteady-state) simulation to analyze the temporal evolution of hydraulic heads and drawdown. Based on the results, a risk analysis scenario is developed, focusing on the system's response to varying pumping rates.

2.1 Model Configuration and Methodology

Building upon the steady-state model developed in the previous tutorial, the setup was modified to accommodate transient conditions.

In the Discretization (DIS) Package, the temporal discretization was refined. Unlike the previous steady-state simulation, this model introduces multiple stress periods to represent changing flow conditions over time. The simulation defines specific lengths for each stress period, setting the status to TR (transient) to capture the storage effects in the aquifer.

Well configuration: The simulation utilizes three pumping wells, maintaining the same spatial characteristics as defined in Point 1.

Output Control and Data Extraction: To capture the dynamic response of the aquifer:

1. Output Control (OC) Package: The code was updated to iterate through all defined stress periods. A loop was implemented to record and save hydraulic head values at every time step (rather than just the final state), effectively creating a "snapshot" of the system at each interval.

Stress Period	Active Wells	Description
Period 1	Well 1 & Well 2	Initial drawdown phase affecting the zones around W1 and W2.
Period 2	Well 2 & Well 3	Well 1 is switched off (recovery), while Well 3 is activated.
Period 3	All Wells	Maximum hydraulic stress scenario with all units pumping simultaneously.

2. Data Extraction: Post-processing was performed using the methodology from Tutorial three. Head and flow data were extracted from the binary output files to allow for time-series analysis.

2.2 Results and Analysis

The drawdown was plotted against time to observe the aquifer's response to pumping.

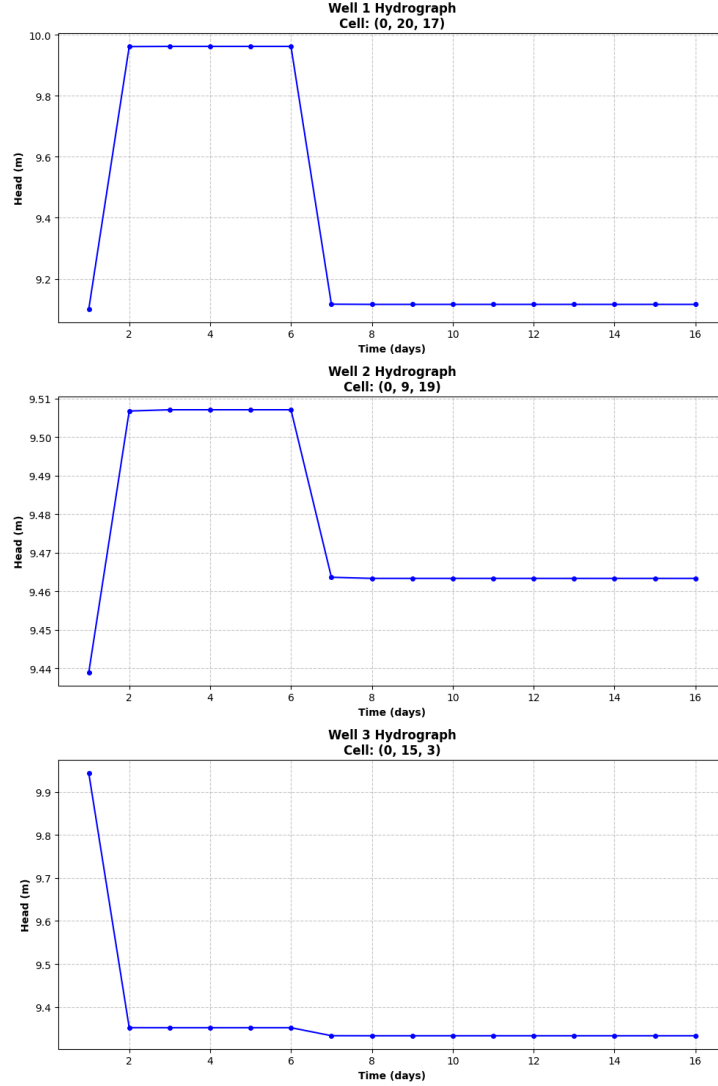


Figure 4: Time-series analysis of drawdown for the three pumping wells. The graph illustrates the rapid decline in hydraulic head during the initial stress periods, stabilizing as the system approaches a pseudo-steady state.

2.3 Risk Analysis Scenario:

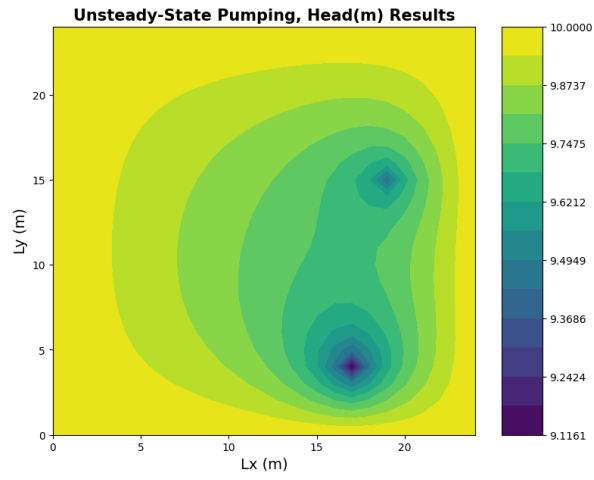


Figure 5: Period 1 (Well 1 Well 2)

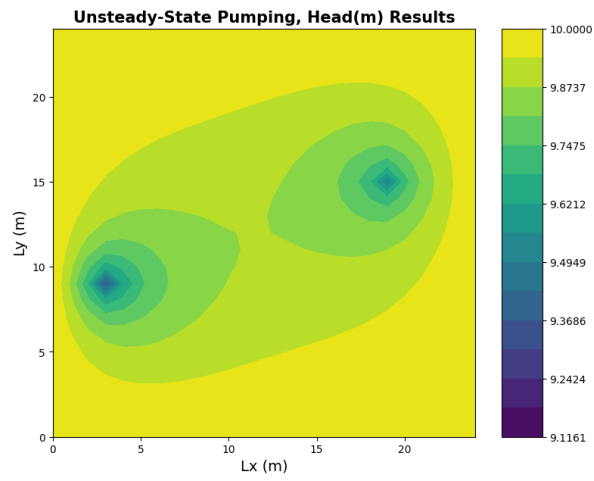


Figure 6: Period 2 (Well 2 Well 3)

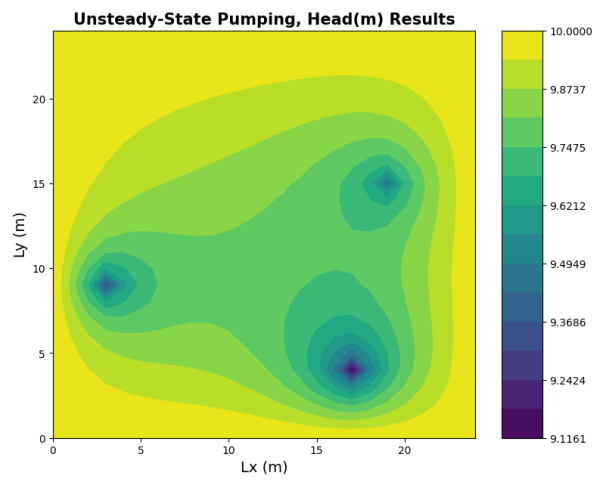


Figure 7: Period 3 (all wells active)

Period 3 represents the "Worst-Case Scenario." The simultaneous operation of all three wells causes the individual cones of depression to intersect (constructive interference). This results in a much deeper drawdown than what would be predicted by summing the individual effects of the wells.

Identified Risks:

- **Maximum Drawdown Limit:** The primary risk occurs in Period 3, where the water level reaches its absolute minimum. There is a potential risk of the hydraulic head dropping below the pump intake or the screen level, which would damage the pumping equipment.
- **Interference Impact:** The reactivation of Well 1 in Period 3, while Well 2 and 3 are already stressing the aquifer, accelerates the depletion rate in the central zone of the well field.

3 Sensitivity analysis to assess the variability of the results with respect to the parameters

3.1 Analysis of point one

To evaluate the sensitivity of the model regarding Point 1, a systematic simulation approach was adopted involving the variation of two key parameters: Hydraulic Conductivity (hk) and Pumping Rates (Q). The analysis was conducted through the following steps:

1. **Base Case Definition:** A reference simulation was run using the calibrated model parameters to establish baseline hydraulic head values.
2. **Scenario Simulation:** Four distinct stress scenarios were generated by modifying the input arrays relative to the base case:
 - Hydraulic Conductivity (hk): Varied by a factor of 50% to represent significant geological uncertainty.
 - Pumping Rates (Q): Varied by a factor of 20% to simulate operational fluctuations.
3. **Data Extraction:** For each of the five total scenarios, the simulated hydraulic heads were extracted at the coordinates corresponding to the three monitoring wells. These values were then compiled to calculate the deviation from the Base Case.

Scenario	Head Well 1 (m)	Head Well 2 (m)	Head Well 3 (m)
Base Case	9.14	9.49	9.77
High K (+50%)	9.43	9.66	9.85
Low K (-50%)	8.29	8.98	9.54
High Q (+20%)	8.97	9.39	9.73
Low Q (-20%)	9.32	9.59	9.82

Table 2: Summary of simulated hydraulic heads (in meters) at the three monitoring wells for the Base Case and the sensitivity analysis scenarios.

Scenario	Head Well 1 (m)	Head Well 2 (m)	Head Well 3 (m)
Base Case	0.00	0.00	0.00
High K (+50%)	0.29	0.17	0.08
Low K (-50%)	-0.86	-0.51	-0.23
High Q (+20%)	-0.17	-0.10	-0.05
Low Q (-20%)	0.17	0.10	0.05

Table 3: Variations in hydraulic head (in meters) relative to the Base Case at the three monitoring wells for the analyzed scenarios.

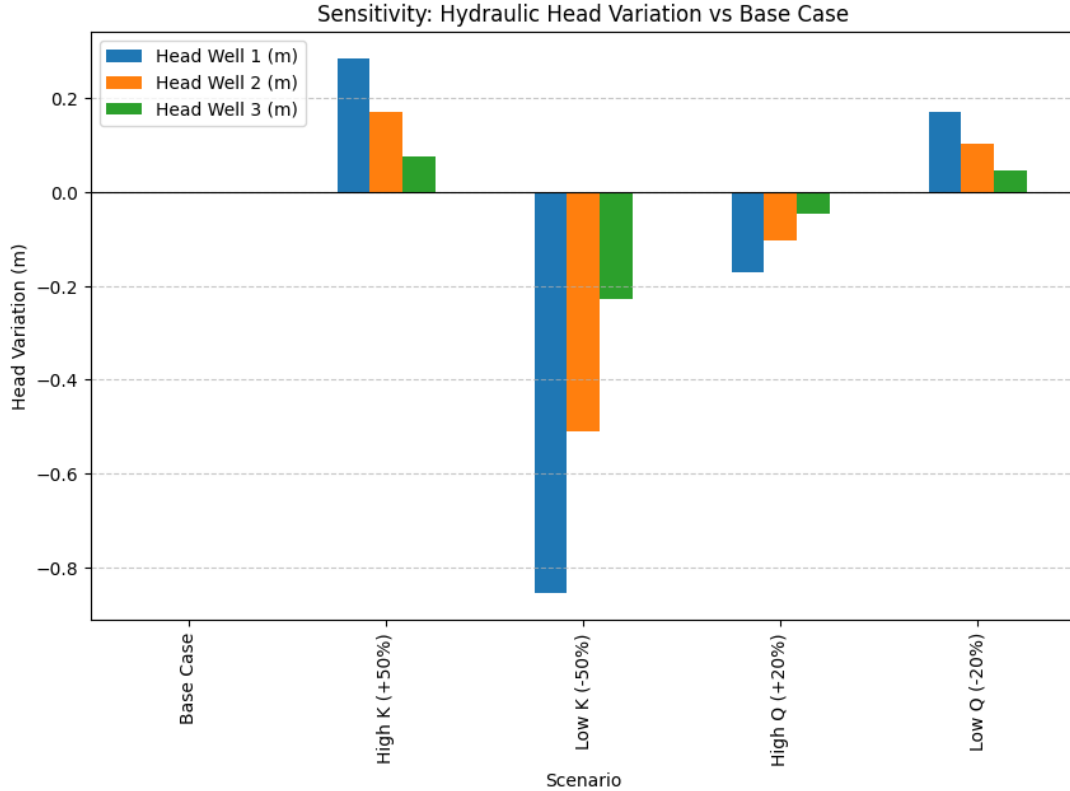


Figure 8: Comparative analysis of hydraulic head deviations at Wells 1, 2, and 3. The chart displays the magnitude of drawdown or recovery under the four stress scenarios compared to the zero-variation base case

3.1.1 Interpretation of the results

The sensitivity analysis was performed to assess how the simulated hydraulic heads in the three wells respond to variations in Hydraulic Conductivity and Pumping Rates Q . The bar chart and the data tables above illustrate the variability of the results under five distinct scenarios:

- **Dominant Influence of Hydraulic Conductivity (hk).** The model exhibits the highest sensitivity to changes in hydraulic conductivity.

1. Low hk Scenario (-50%): Reducing hk resulted in the most significant draw-down (largest negative variation in head). This indicates that when the aquifer

material is less permeable, the hydraulic gradient must steepen significantly to sustain the prescribed pumping rates, causing a deeper cone of depression.

2. **High hk Scenario (+50%):** Conversely, increasing hk allows water to flow more easily toward the wells, resulting in higher hydraulic heads (less drawdown) compared to the base case.
- **Impact of Pumping Rates (Q):** the results show a linear and expected response to pumping stress, though the magnitude of impact is lower compared to the changes in hk (for the tested ranges).
 1. **High Q (+20%):** Increasing the extraction rate leads to a moderate decline in water levels.
 2. **Low Q (-20%):** Decreasing the extraction rate results in a recovery of water levels.

So, at the end we can say that the analysis demonstrates that the model results are highly sensitive to the estimation of Hydraulic Conductivity. Therefore, accurate field characterization of hk is critical for the reliability of this model, as errors in this parameter will lead to the largest deviations in predicted water levels.

3.2 Analysis of point two

Following the risk analysis and the definition of the conceptual model, a sensitivity analysis was performed to assess how variations in stress period duration and pumping rates affect the response of the hydrogeological system. Given the transient nature of the simulation, these parameters are critical in determining the magnitude and extent of piezometric drawdown.

3.2.1 Sensitivity to Stress Period Duration

The model is configured in a transient regime for all three simulation periods. This implies that the system does not necessarily reach equilibrium (steady-state) in each phase, making the “time” factor a fundamental variable.

1. **Short Periods (e.g., Stress Period 1 – 1 day):** The brevity of the first period limits the expansion of the cone of depression generated by Wells 1 and 2. An increase in the duration of this period would result in greater initial drawdown, altering the initial conditions for subsequent periods. Consequently, the system is highly sensitive to this parameter: if the period is too short, there is a risk of underestimating the initial impact of the wells.
2. **Long and Cumulative Periods (e.g., Stress Period 3 – 10 days):** The third period is the longest and sees all three wells active simultaneously. As this is the phase of maximum hydraulic stress, sensitivity is at its peak here: extending the duration would lead to a further expansion of the cone of depression toward the domain boundaries. If the period is not sufficiently long, the model may fail to capture the maximum potential drawdown before the end of the simulation.

So, at the end, the analysis confirms that in a transient simulation, the duration of stress periods acts as a primary sensitivity parameter. Unlike steady-state simulations where the system reaches equilibrium regardless of time, here the hydraulic response is cumulative and time-dependent.